Lecture 3

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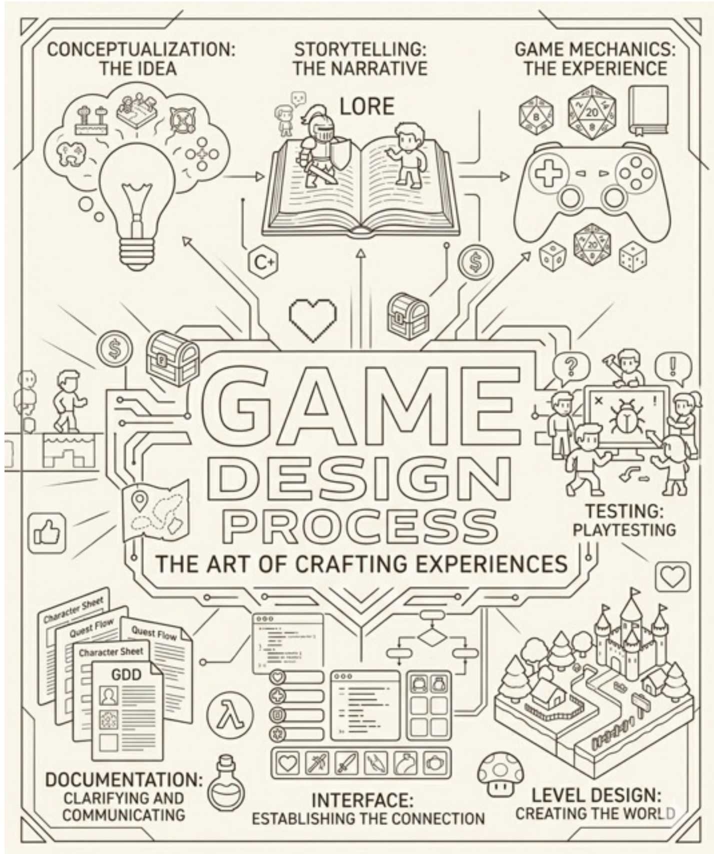
Game Design
(cont.)

Game Design

- Conceptualization: The Idea
- Storytelling: The narrative
- Game mechanics: the Experience
- **Level Design: Creating the World**
- Interface: Establishing the Connection
- Documentation: Clarifying and Communicating
- Testing: Playtesting

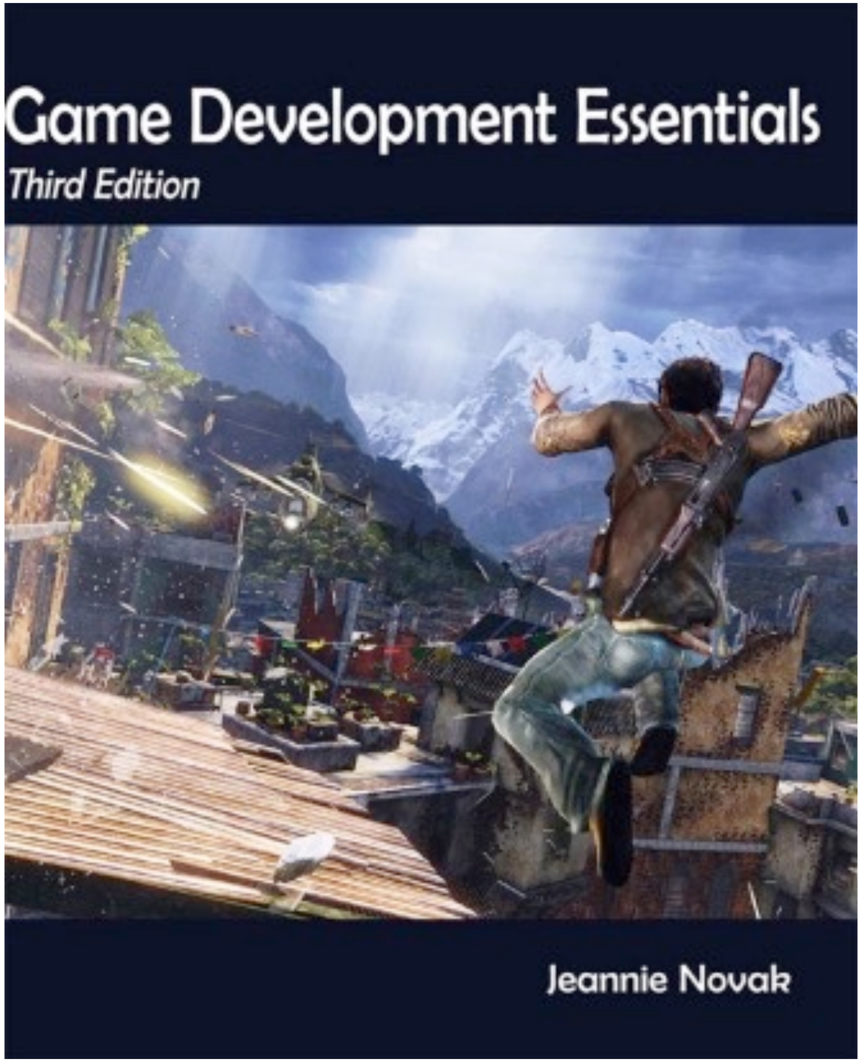


Level Design

References

Game Development Essentials

3rd edition
Jeannie Novak, 2012
ISBN-13: 978-1111307653



Level Design

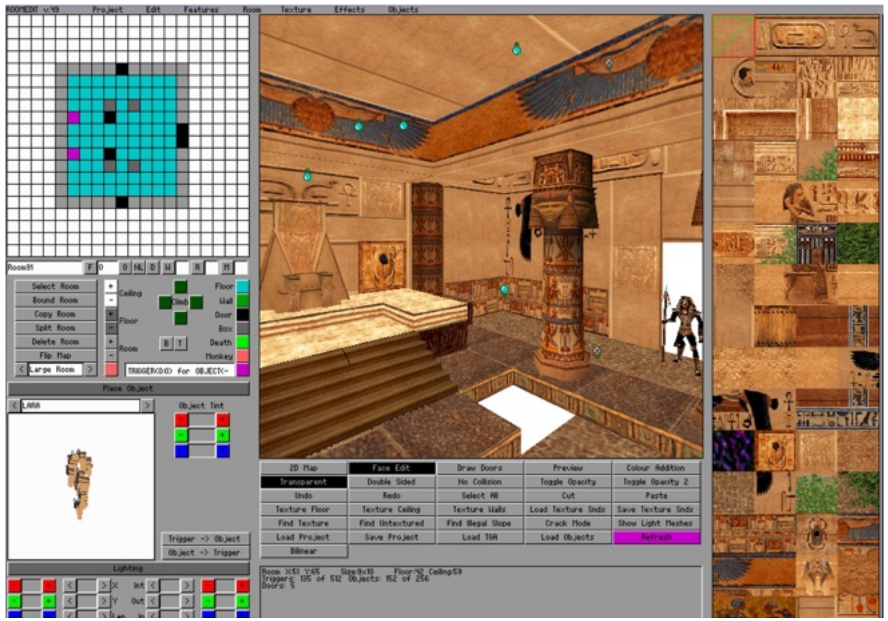
- "*Level Design is defined as the creation of environments, scenarios, or missions in an electronic game.*"

  In Jeannie Novak, "Game Development Essentials"

- Introduces new characters or objects
- Focus on a plot point (such as discovering a secret or preventing an attack)
- Create a mood through visuals or storyline
- Centers around an idea that becomes a unifying theme

Level Design

- Level Design Tools: Limited set of options within the game
  - Ex: Valve Hammer Editor, EA World Builder
- Game Engines: Generic unlimited possibilities of editing
  - Ex: Unity 3D, Unreal Engine
- 3D modeling and media authoring tools: specialized tools to edit 3D meshes, character skeletons and animations, textures, audio
  - Ex: 3DS Max, Maya, Blender, Audacity

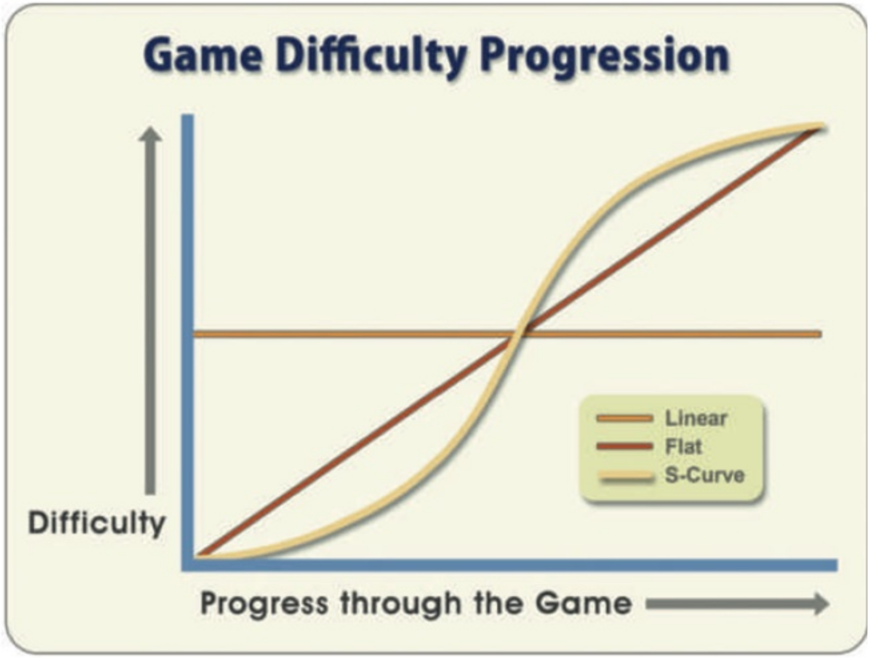


[Tomb Raider level editor, 1996]

Structure

- Levels structure a game into effective subdivisions, organize progression and enhance gameplay.
- When creating levels consider:
  - Previously defined goals according to story
  - Flow of the mechanics and story: direct player to follow the main plot
  - Duration (mini game 15 min, regular console game 45 min of a 60 hours game)
  - Availability: How many levels? How many simultaneous paths?
  - Relationship with story and gameplay.
  - Difficulty: how difficult is the game progression?

Diagram by Per Olin



Comparison of linear, flat, and s-curve progressions.

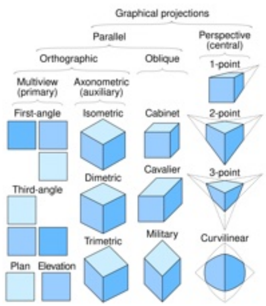
Time

- “Game Time” can be slower, faster or real-time:
  - **Authentic**: time is the real-world time
    - Ex: World of Warcraft
  - **Limited**: time is part of the mechanics
    - Ex: gameplay is limited for some time as in FarmVille
  - **Variable**: when it is not important, time goes faster
  - **Limited-Player-Adjusted**: the player can adjust the total time
    - Ex: sports games
  - **Altered**: The player can slowdown time to perform a task
    - Ex: Prince of Persia or Max Payne

[Max Payne slowing Time]

Space

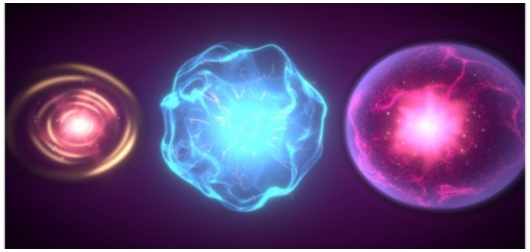
- Physical environment of the game - including its perspective, scale, boundaries, structures, terrain, objects, style and lighting.
- Camera
    - Position and orientation
        - Aerial (Top-Down)
        - Side-scrolling (Flat / Side view)
    - Projection type:
        - Orthographic
        - Perspective

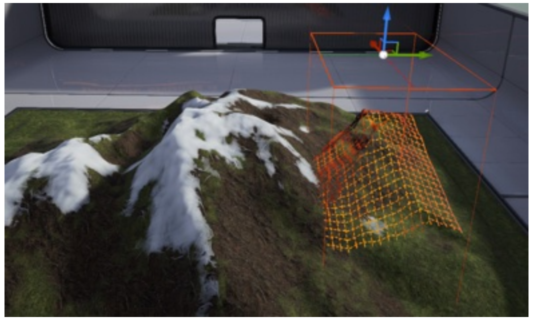


[StarCraft, 1998]

Other Aspects

- Things to consider while creating a game level
  - Terrain and Materials
  - Visual Effects
  - Lighting
  - Audio
  - Character Animation





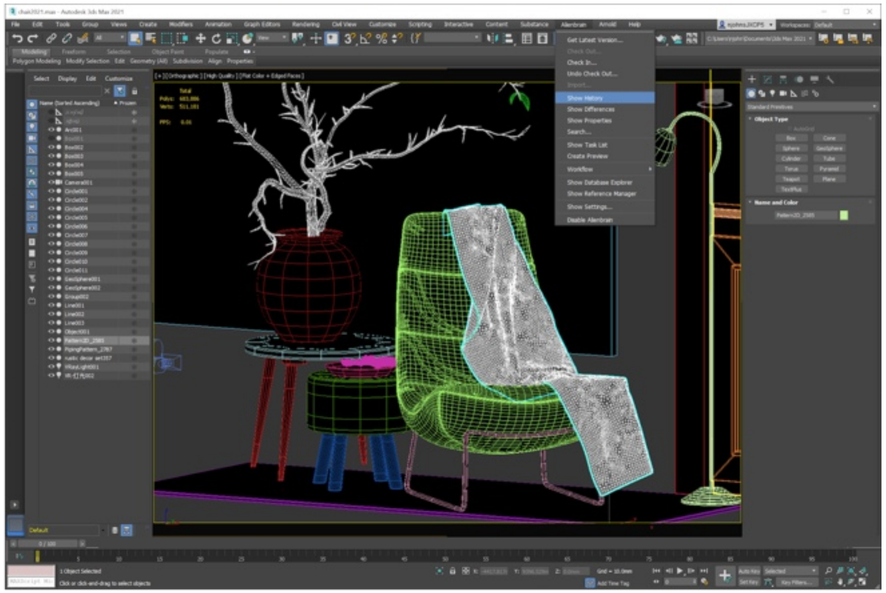
3D Modeling

- 3D modeling tools allow the creation of assets: objects, structures and worlds
- Main file formats for 3D meshes:
  - Autodesk **.3ds**, **.fbx**, **.dwg**
  - Kronos Group COLLADA **.dae**
  - Wavefront **.obj** and 3D Systems **.stl**
  - Web3D **.wrl**, **.x3d**
  - Blender **.blend**

Format	Textures / PBR	Animation	Native LODs	Streaming Adequacy	Encoding	Compression
.glb	**Yes (Full)**	Yes	**High** (MSFT_lod)	**Excellent**	Binary	**Draco**
.gltf	**Yes (Full)**	Yes	**High** (MSFT_lod)	**Excellent**	Text (JSON)	**Draco**
.fbx	Yes	Yes	**High** (Groups)	**Moderate**	Binary	None
.blend	Yes	Yes	Internal Only	**Poor**	Binary	Zstandard
.obj	Yes (Legacy)	No	**None**	**Poor**	Text (ASCII)	None
.dae	Yes	Yes	Partial (XML)	**Poor**	Text (XML)	None
.wrl	Yes	Yes	Yes (Nodes)	**Moderate**	Text (ASCII)	Gzip
.3ds	Yes (Legacy)	Basic	**No**	**Poor**	Binary	None
.stl	No	No	**No**	**Poor**	Binary/Text	None
.dwg	Yes	No	BIM Level	**Poor**	Binary	Internal

Autodesk

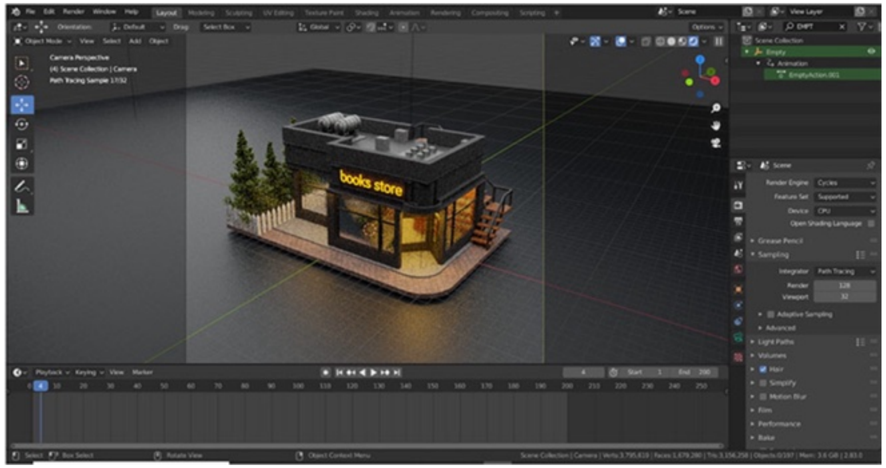
- Autodesk specializes in 3D modeling offering a variety of software with different specialties:
    - 3ds Max: mesh modeling
    - Maya: cinema and rendering
    - Fusion 3D: 3D printing
    - AutoCAD: architecture
    - Recap: photogrammetry



[Autodesk 3D Studio Max]

Blender

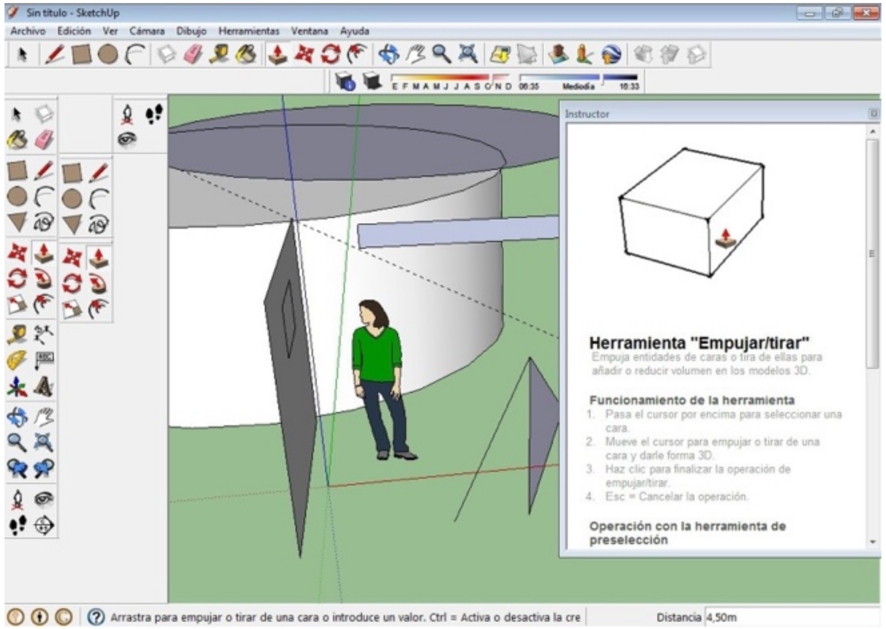
- Blender is an open-source alternative capable of mesh modeling, rendering and animation
    - Python scripting language
    - Character animation plugins
    - Integration with several pipelines



[Blender]

SketchUp

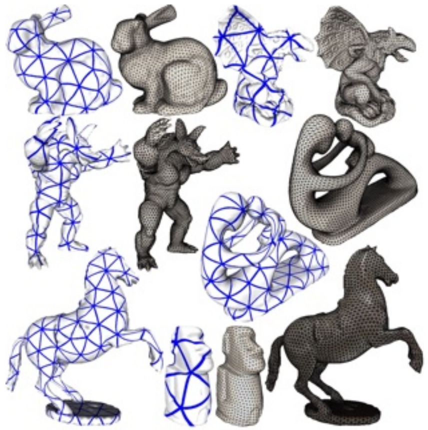
- Sketch based modeling
- Initially used by Google to create assets for Google Earth
- Fast development of simple models
- Now owned by Trimble

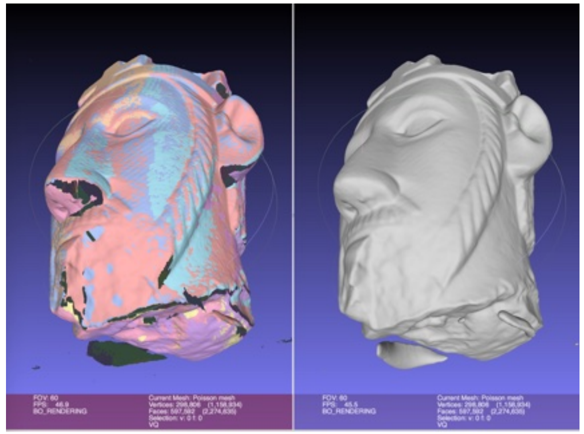


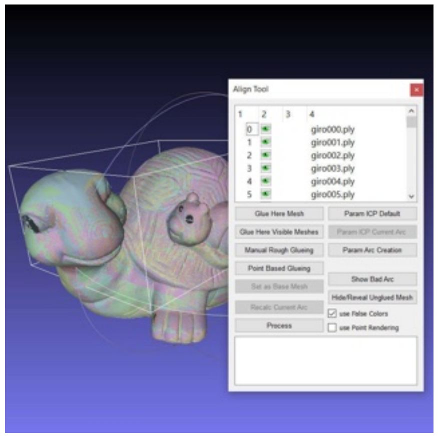
[SketchUp 8]

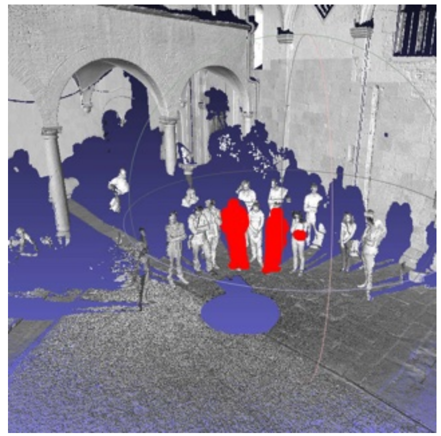
MeshLab

- 3D Mesh Processor and Editor
- Supports multiple formats
- Used for:
    - 3D Acquisition (aligning, reconstruction, color mapping and texturing)
    - Clean, compare, convert, simplify, refine and remesh 3D meshes
    - Measurement and Analysis









Character Animation

- **Standard 3D Pipeline** (Keyframes & MoCap)
  - High-end film and AAA gaming.
  - **Keyframe animation**: Manual control of every pose.
  - **Motion Capture**: Recording an actor's movement (marker vs. marker less based)
  - **Hybrid approach**: Mocaps as base layer and animators keyframe over it to add "appeal" or fix "uncanny valley" jitters.

Character Animation

Software	Best For	Key Highlights
Autodesk Maya	Professional 3D	The industry leader for rigging and complex character movement. The 2026 update features **MotionMaker**, a generative system for natural movement.
Blender	All-in-One (Free)	Completely free and open-source. Version 4.3+ has a massive **Grease Pencil v3** update, making it the best tool for "2D-in-3D" hybrid styles.
Toon Boom Harmony	Professional 2D	The go-to for TV shows like *The Simpsons*. It offers the most advanced paperless and cut-out animation rigging available.
Cinema 4D	3D Motion Graphics	Known for its intuitive UI and seamless integration with After Effects. Popular for "designer-friendly" 3D.

Character Animation

- **Real-Time pipeline** (Engine first workflow)
    - Instead of waiting hours to render a scene studios are moving production into game engines.
    - **Virtual production**: Animators and directors work inside a VR headset or on an LED volume stage, seeing the final lighting and character movements live as they happen.
    - **Live Link**: Using an iPhone or specialized camera to stream facial expressions directly onto a 3D character in real-time—extremely popular for VTubers and social media content.

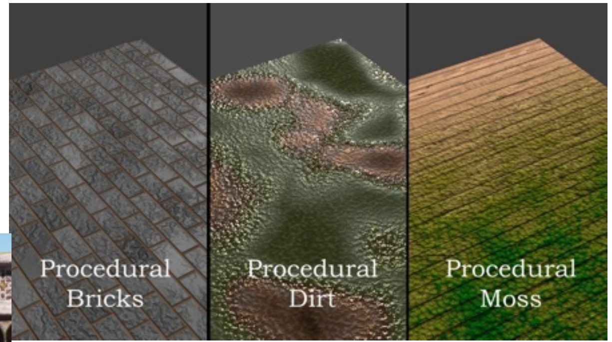
Character Animation

- **AI-Assisted & Generative Pipelines** (The New Alternative)
- AI is no longer just for "generating images"; it's a core part of the animation toolset.
    - **Automated In-Betweening**: AI tools (like those in Adobe or Toon Boom) now handle the tedious "in-between" frames, allowing artists to focus only on the main keyframes.
    - **Text-to-Motion**: New models (e.g., Bumblebee's Motifect) allow you to type "character walks sadly" to generate a game-ready 3D animation sequence instantly.
    - **Character Consistency Engines**: Tools like FLUX.1 Kontext ensure that a character's design stays identical across thousands of frames without manual "cleanup."

Procedural Content Generation

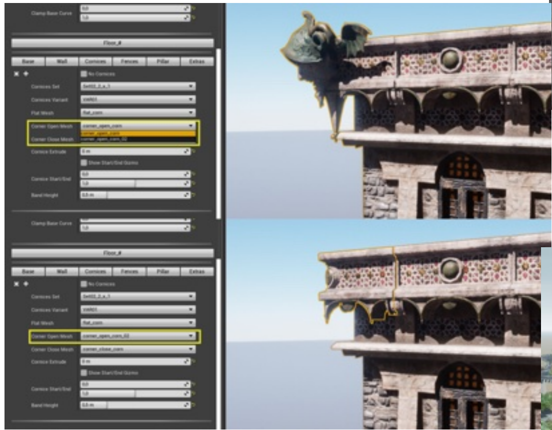
- Advantages
  - Occupies less space
  - Scales easily according to the rules
  - Good for fictitious locations
- Disadvantages
  - More processing time
  - Tends to have some repetition
  - Not good for known locations

Materials



[from Reddit discussion]

Buildings



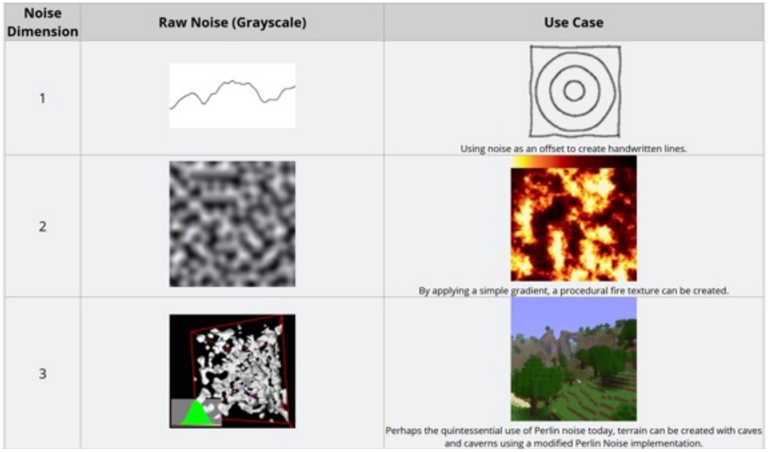
[Houdini Building Generator for UE]

Cities

[Ultimate City Generator]

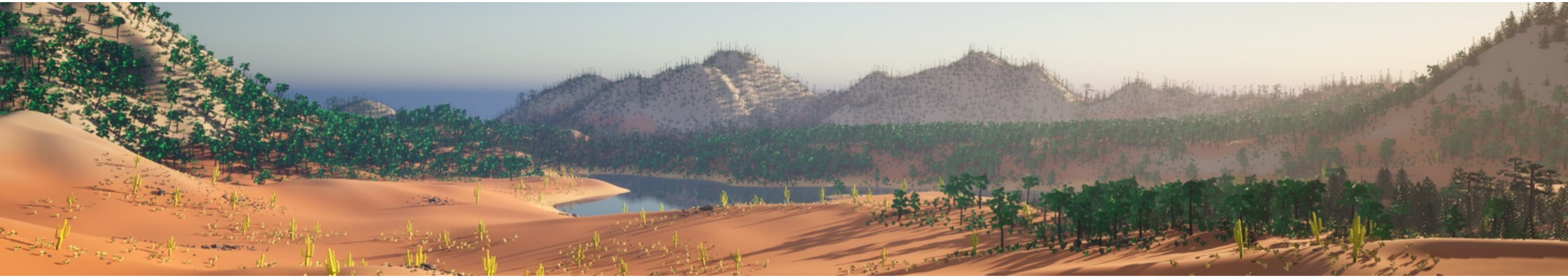
Terrain Generation

- Noise can be used to create natural and organic features.
- Using Perlin Noise it is possible to create a pseudo-random realistic terrain.



Noise Dimension	Raw Noise (Grayscale)	Use Case
1		Using noise as an offset to create handwritten lines.
2		By applying a simple gradient, a procedural fire texture can be created.
3		Perhaps the quintessential use of Perlin noise today, terrain can be created with caves and caverns using a modified Perlin Noise implementation.

[Adrian's soapbox]



[Unreal PCG Biome Core]

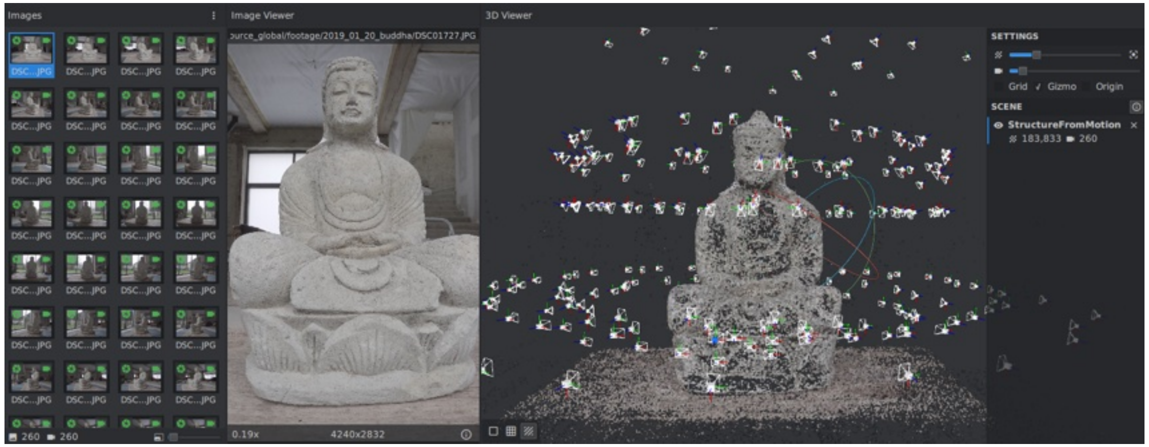
Photogrammetry

- Photogrammetry is the use of photography in surveying and mapping to ascertain measurements between objects.
- Using several photos it is possible to do 3D reconstruction using photogrammetry.
- Used in games to create realistic natural worlds, natural textures or realistic scenes from the real world.

[Palácio da Pena, Sintra, Google Earth]

Photogrammetry

- Pros: reduces creation time
- Cons: manual post-processing and cleaning
- Software:
  - Autodesk Recap
  - Agisoft Metashape
  - AliceVision Mushroom
  - Epic Reality Scan
  - Kiri Engine
  - Matterport

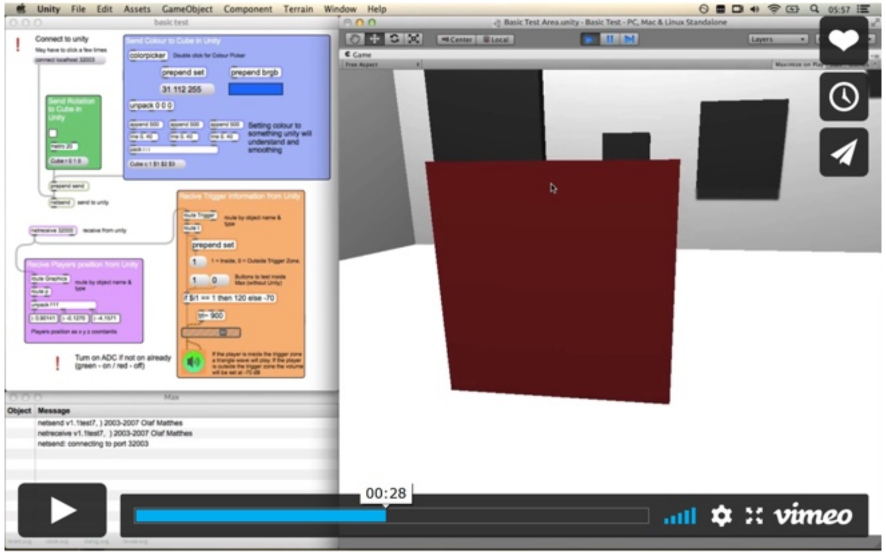


[AliceVision Meshroom]

[PhotoTourism: Building Rome in a Day, 2009]

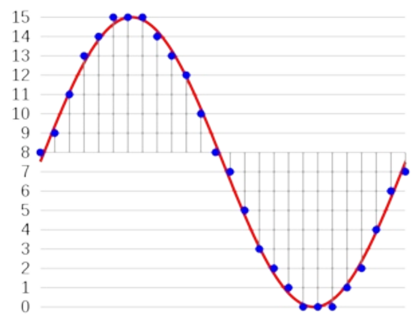
Audio

- Used in background music and sound effects (SFX)
- **Event-driven**: something happens, sound is played.
- **Procedurally Generated**: Audio depends on game parameters such as steps or how much the door is open.
- Applications: Audacity, Adobe Audition, Reaper, Max MSP



[Max MSP + Unity]

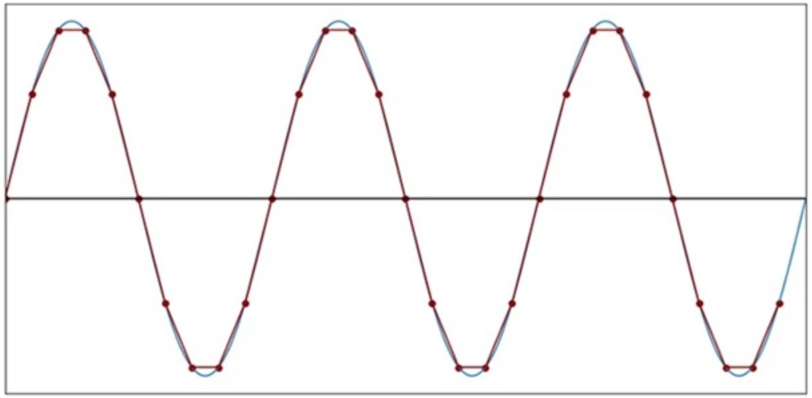
Audio



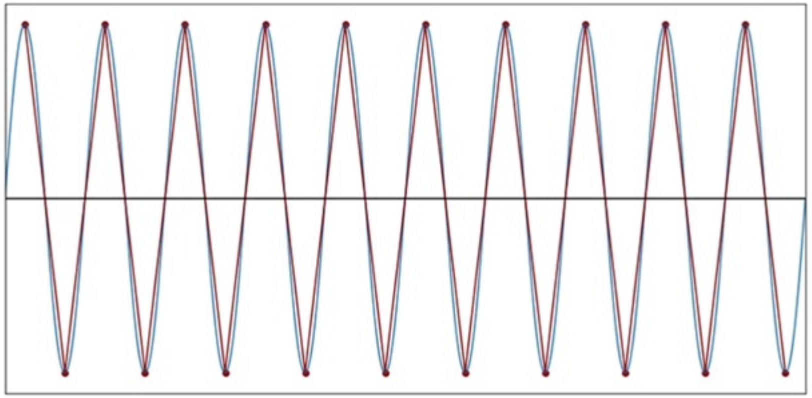
[4 bit sampling of a 1Hz wave at 26 Hz]

- **Nyquist-Shannon sampling theorem**
  - If a function $f(x)$ contains no frequencies higher than B Hz, it is completely determined by giving its ordinates at a series of points spaced less than $1/2B$ seconds apart.
  - $f = 1/T$ frequency is measured in Hertz (Hz), it is the number of times there is a full wave in one second. T is the period, the time of each wave.
  - There needs to be 2+ samples per complete wave to reconstruct the original signal.
  - Humans can hear up to $22KHz$, CD recordings are samples at $44.1KHz$

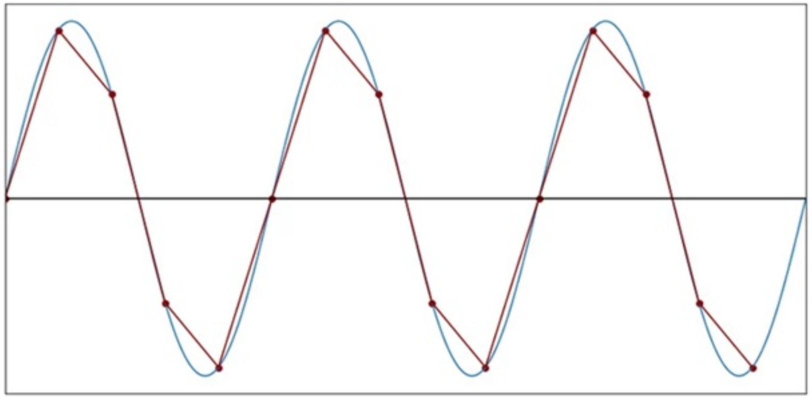
Audio



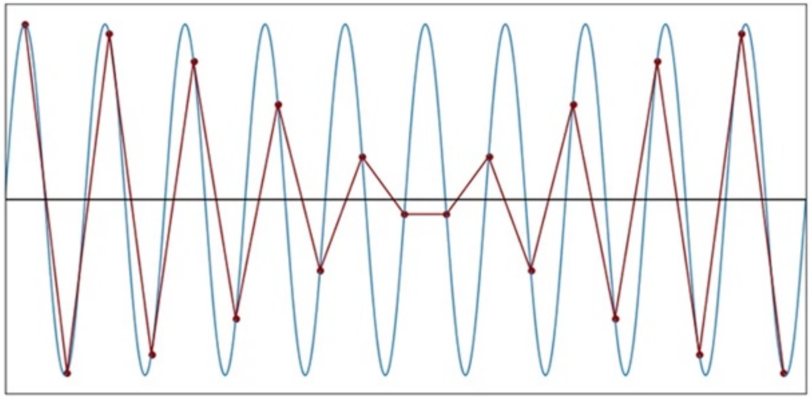
10 samples per cycle



2 samples per cycle



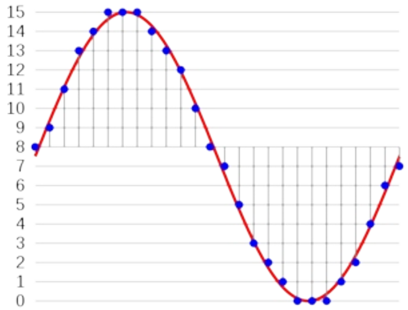
5 samples per cycle



1.9 samples per cycle

Fernando Birra
Rui Nóbrega

Audio



[4 bit sampling of a 1Hz wave at 26 Hz]
26 samples x 1/2 byte x 1 channel = 13 bytes

- Digital audio is composed of
  - N samples per second (frequency)
  - Each sample occupies b bits of space
  - Sound can be replicated in several channels
- PCM (pulse-code modulated) audio (.wav, cd audio)
  - Frequency: 22 KHz, 44 KHz
  - Bits per sample: 8, 16, 24 bit
  - Channels: mono, stereo, ambisonics (4 channels), 5.1 Surround
- CD Audio: 44.1 KHz, 16 bit, stereo > 1 s of audio is 44100 (samples) x 2 (bytes) x 2 (channels) = 172 KB
- MP3 Audio: lossy compression, frequency domain representation, psychoacoustics → much less space occupied.

Images

- Used to create Textures and Sprites
  - Resolution: usually square powers of 2 (256x256, 512x512, …)
  - Color Mode: defines the number of channels and color function
  - Color Depth: defines the bits per channel, usually 8 bits. HDR uses 16 bit. RGBA is a 4-channel mode with 32-bit color depth.
  - DPI: dots per inch, relation between pixels and real-world measurements. Mainly relevant for printing or relating with screen monitor size
    - **8 bits per channel** - Grayscale: 8 bits per pixel (1 byte) , RGB: 24 bits per pixel (3 bytes) , RGBA : 32 bits per pixel (3 bytes), HSV: 24 bits per pixel (3 bytes).
    - **Custom** - 16 colors: 4 bits per pixel, B/W: 1 bit per pixel
  - Formats:
    - **No compression**: .bmp, .raw
    - **Lossless compression**: .png (lossless mode), .tiff, run-length encoding methods
    - **Lossy compression**: .jpg, .png, .gif (8 bit color)

Pixel Art

- Sprites are rectangular segments of a texture
- Pixel Art is the art of designing objects with a limited number of pixels and colors
- Makes use of:
    - Exaggerated features
    - Defined outlines
    - High-contrast colors
    - Dithering instead of gradient colors

[Super Mario World]

Game Design

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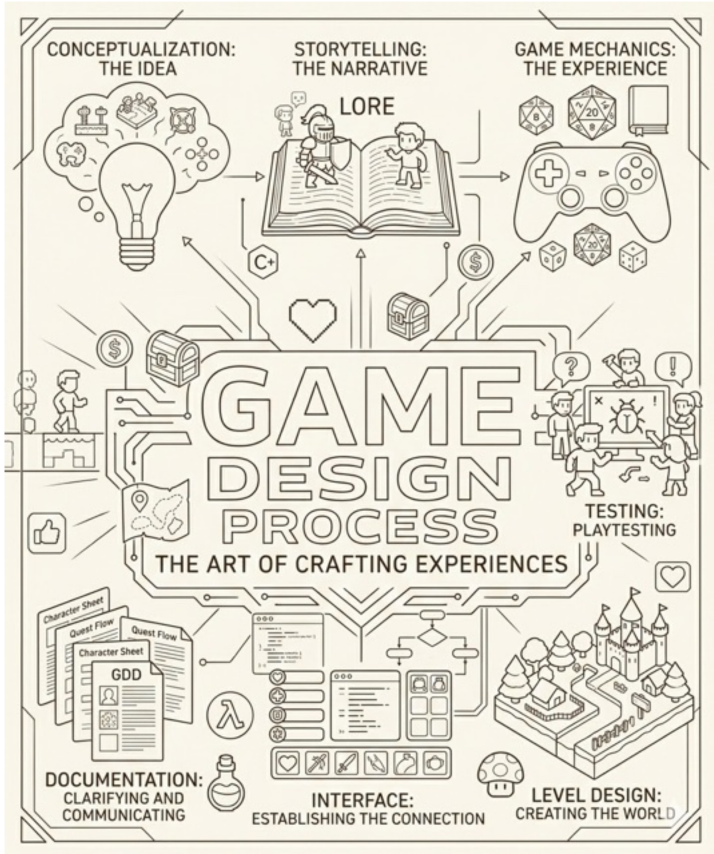
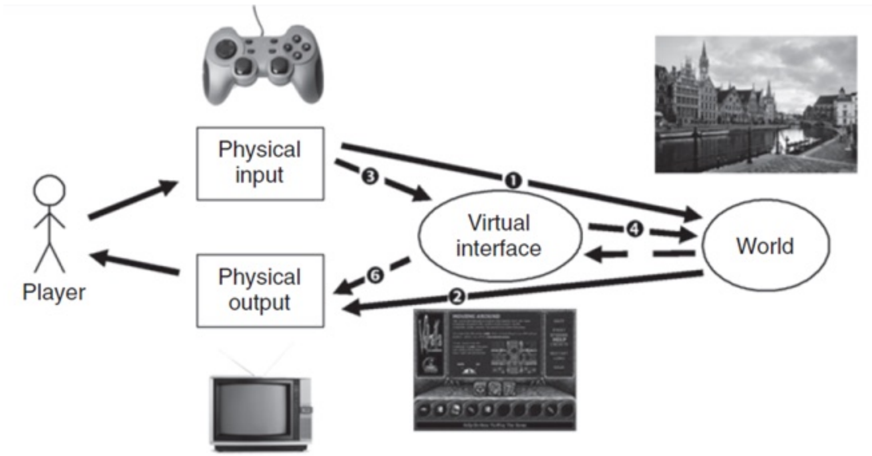


Interface

Interface Design

- The interface is the connection between the player and the game
  - Physical Interface (input): gamepad, keyboard, mouse, joystick, etc.
  - Visual (Virtual) interface: Buttons, Billboards, Visual Information, Diegetic vs non-diegetic information, etc.
  - The interface provides input but also sensory feedback (output)



Interface Transparency

- Does the interface let the players do what they want?
- Do new players find the interface intuitive?
- Can players learn to use it without thinking?
- Is the interface over-confusing?
- Can players continue to use the interface well in stressful conditions?
- Does something confuse the players about the interface? On which of the six interface arrows is it happening?

Player-Centered Design

- Modified version of user-centered design
  - You should create an interface for something because the player needs it…
  - … not because the system may provide it!

[Character Personalization, Lord Of the Rings]

[Black & White 2, the interface uses mouse gestures and no buttons]

Physical Interface

- Arcade Games: custom physical interface for each game
- Computer Games: mouse and keyboard available
- Console Games: gamepad
- Mobile Games: Touchscreen
- VR: Headset and wands
- Motion sensing: Gestures
- Others: Balance boards, dance carpets, …

Visual Interfaces

- **Active**: Players can interact - menus, options, customizations, buttons, sliders, etc.
- **Passive**: Player cannot interact - screen displays, informational panels and billboards, status, goal information, etc.
- Typical components: score, lives & power, map, character config, start screen, status panels, goals list, inventory, simulation panel, etc.
- Audio is sometimes considered to be a passive interface.

Visual Interfaces

- **Visible**: Players can see the interface - menus, options customizations, buttons, sliders, etc.
- **Hidden**: Player has to know the interface - keyboard keys, mouse gestures, swipe gestures.
    - May require tutorials for the player to learn!

[Street Fighter IV, 2008]

Visual Interfaces

- Diegetic vs non-Diegetic
- A Diegetic interface is when a game's interface elements exist inside the World
- Examples from: Diegesis Theory

[Far Cry 2, diegetic]

[World of Warcraft, non-diegetic]

[Far Cry 2, diegetic]

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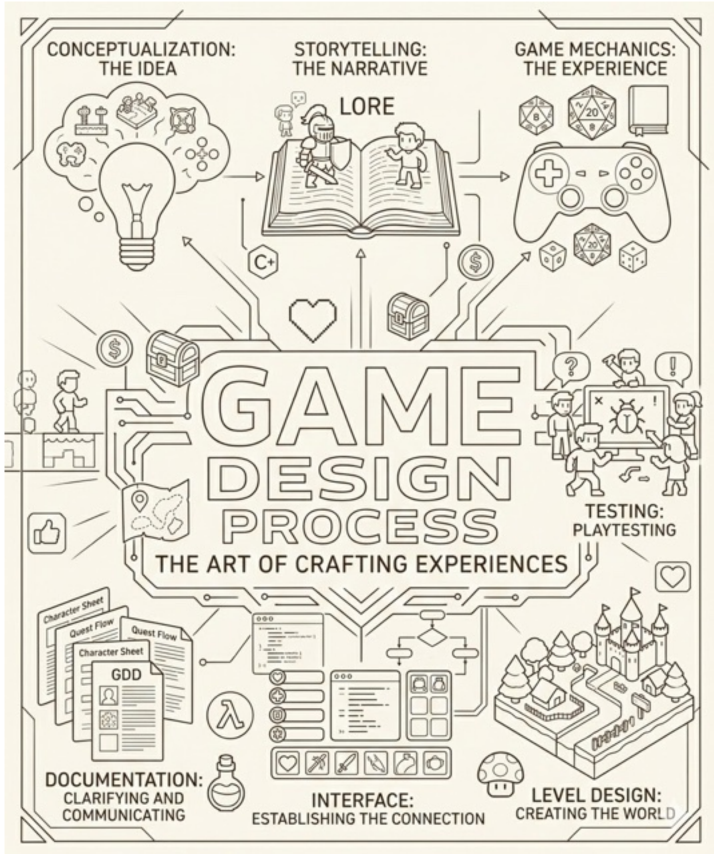

Documentation

Documentation

- Documentation is created along the development
- Main purposes:
    - Ensure that team members understand their roles
    - Convince companies to invest/fund development
    - Document for future developers
- No standard exists!

Concept Document (I)

- Convey the goal and purpose of the game
  - Premise (high concept): The summary of the game purpose (1-2 sentences)
  - Player (the character) motivation: The game's victory condition. What will drive the player to keep playing the game to the end.
  - Unique selling proposition (USP): Why would anyone choose this game over competition.
  - Target market: The portion of the game-playing audience the game is aimed at.
  - Genre: What will be the primary genre for the game?

Concept Document (II)

- Target rating: age target, PEGI / ESBR rating for the game.
- Target platform and HW requirements: Choose the platform(s) for the game.
- License: Licensing information.
- Competitive analysis: Select 5 successful titles for a comparative analysis.
- Goals: What are the expectations for this game as an experience? Discuss how the game will achieve these goals.

Game Proposal

- Describe the components of the concept with further detail. Include:
  - Title, Hook, Goals
  - Story Synopsis, Backstory, Characters
  - Gameplay and Mechanics
  - Technology, Online Features
  - Art and Audio Features, Concept Art
  - Production Details (Team, roles, budget, schedule)
  - Risk Analysis, Marketing, intellectual property

Game Design Document

- The large reference guide to the development process with a definitive version of what was proposed in the Game Proposal
  - Story: Story Bible
  - Game interface
  - Character Abilities and Items
  - Game World: Game Level Design for each level
  - Game Engine: Technical Design Document, Diagrams, Technical drawings and schematics
  - Concept Art
  - Project Plan: Team, Roles, budget, schedule, milestones
  - Test Plan